

Mixture of Experts

DATA 443/643 · Week 7, note 7.2

Week 7

American University

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Ideas from Stanford CS336 2025 L4 (Mixtral: 8 routed, top-2 active). Original slides; that course is not copied.

Lecture plan

- 1 Router, then k experts
- 2 Stored weights versus active FLOPs
- 3 Load collapse
- 4 When a dense 7B is the better project

Key goal: 8 MLPs on disk, 2 in the FLOP count. Say that out loud.

First time: what / why / formula

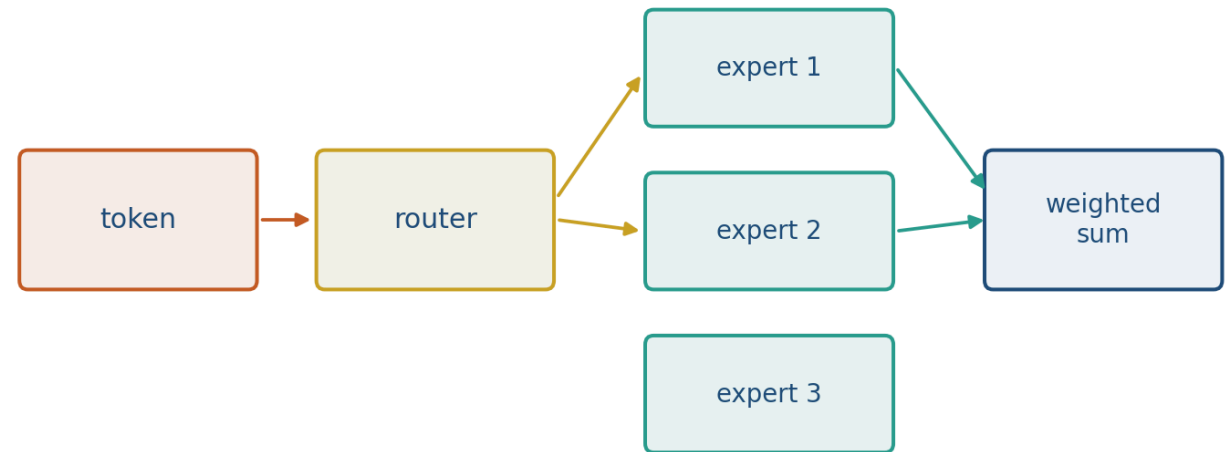
- What / why. A router picks k experts (often 2 of 8). Capacity grows with stored experts; FLOPs stay near k dense MLPs. Dense nets pay every MLP on every token. MoE buys width in parameters without width in compute.
- Architecture. Attention stays dense. Replace the block MLP with E copies plus a linear router. Mixtral cartoon: 8 stored, top-2 active, no shared expert.
- Formula. $y = \sum_{i \in \mathcal{T}(x)} g_i(x) E_i(x)$. Stored MLP weights $\approx EP$; active FLOPs scale with k .
- Tradeoffs. + More capacity at similar token-FLOPs. – RAM holds all experts; batching is harder; collapse = a dense net in disguise; serving $\neq$ “same active FLOPs.”

How. Softmax (or noisy top- k) over router logits, run those experts, weighted sum. Load-balancing so expert 0 does not eat the batch.

Sparse where the parameters already live

- Attention can stay dense. The MLP is the MoE.
- CS336: Mixtral is 8 experts, $k=2$. Switch used $k=1$.
- DeepSeek's 256-expert nets wait. This hour is the picture.

Sparse MoE: most experts stay off for a given token



Top-k mix

$$y = \sum_{i \in \mathcal{T}(x)} g_i(x) E_i(x)$$

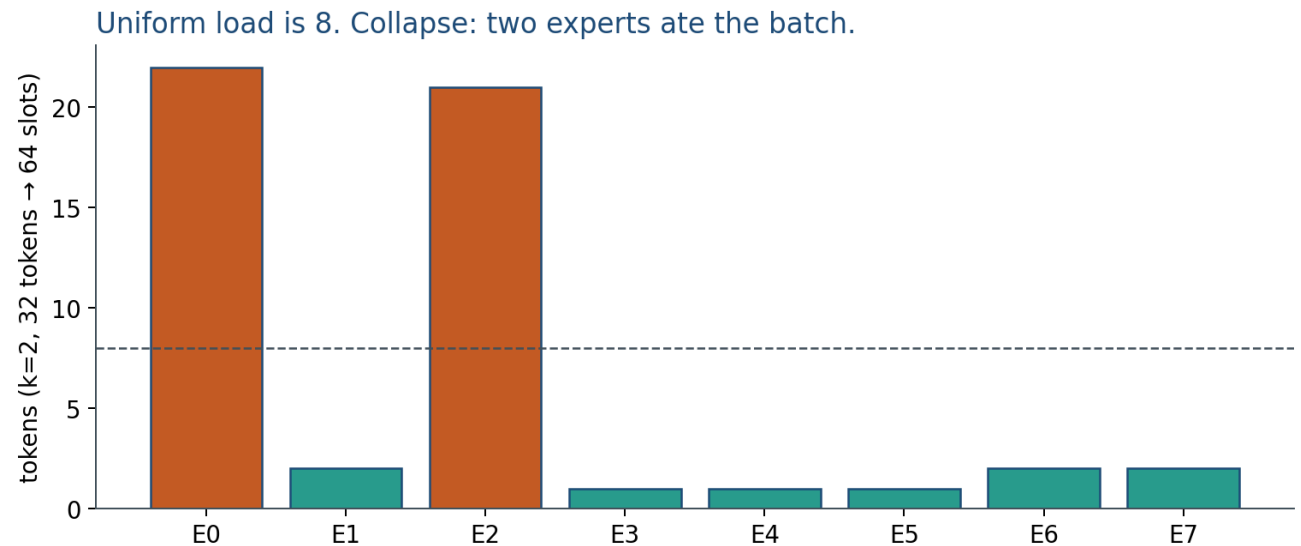
$\mathcal{T}$ is the k selected experts.

Renormalize the two gates so they sum to 1.

If every token has the same logits, you trained two MLPs.

Collapse is a dense net in disguise

- 32 tokens, $k=2$: 64 slots. Uniform load is 8 per expert.
- If two experts eat 43 slots, six experts are cold.
- Load-balancing tries to stop that. Token drop is the other knob.



What you actually ship

Dense 7B

One weight file, a dense GEMM, simple batching.

Mixtral-style 8x

Eight MLP shards in RAM even if two run. Tokens in one sequence may disagree on the expert.

What to take from this lecture

Stored $\neq$ active

8× MLP weights, $\sim 2\times$ MLP FLOPs.

Router

If it always picks expert 2, you trained one expert.

Lab 7

Scaling fit and INT8, not an MoE train.